

# EDV152-电动床防夹

使用说明书 / User Manual

深圳市易探科技有限公司

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外部版 / External Version

# 版本记录 / Revision History

| 序号<br>No. | 版本<br>Version | 变更说明<br>Description                                                                                                  | 日期<br>Date | 作者<br>Author |
|-----------|---------------|----------------------------------------------------------------------------------------------------------------------|------------|--------------|
| 1         | V0.1          | 创建文档<br>Create document                                                                                              | 2026/5/6   | 严哲峰          |
| 2         | V0.2          | 补充关于主从模块通信的说明<br>Supplement communication instructions<br>for master-slave modules                                   | 2026/6/4   | 严哲峰          |
| 3         | V0.3          | 补充使用说明<br>Supplement usage instructions                                                                              | 2026/6/6   | 严哲峰          |
| 4         | V0.4          | 改为整套设备说明<br>Changed to complete equipment<br>description                                                             | 2026/6/25  | 严哲峰          |
| 5         | V0.5          | 1 修改相应的实物<br>2 补充使用说明<br>1 Update corresponding product photos<br>2 Supplement usage instructions                    | 2026/7/18  | 严哲峰          |
| 6         | V0.6          | 1 定义检测区域和保护区域<br>2 确定区域相关指标<br>1 Define detection zone and protection<br>zone<br>2 Determine zone related indicators | 2026/7/25  | 严哲峰          |

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## 第一部分 / Chapter 1

# 产品介绍 / Product Introduction

### 1.1 产品概述 / Overview

EDV152是一款工作在60GHz频段的毫米波雷达，主要用于检测人体运动/存在状态，在此基础上做到了更高的距离精度、响应速度以及角度控制，可用于窄角度占位应用、智能设备联动等场景。

EDV152 is a millimeter-wave radar operating in the 60GHz frequency band, mainly used to detect human movement/presence. It achieves higher distance accuracy, response speed and angle control, and can be used in narrow-angle occupancy applications, smart device linkage and other scenarios.

在智能床防夹场景下，需要成对使用：其中1个作为主模块，负责与客户上位机交互，并转接从模块。主模块可以被配置为左侧或者右侧。主模块通过485协议与客户上位机通信。

In the smart bed anti-pinch scenario, it needs to be used in pairs: one as the master module, responsible for interacting with the customer's host computer and forwarding the slave module. The master module can be configured as left or right side. The master module communicates with the customer's host computer via RS485 protocol.

系统上电后持续上报如果有活动目标进入床周边约20厘米范围，主模块会向客户上位机报告入侵事件。

After the system is powered on, it continuously reports. If an active target enters the area within about 20cm around the bed, the master module will report an intrusion event to the customer's host computer.

### 1.2 产品特点 / Features

- 支持人体运动/存在检测  
Supports human movement/presence detection
- 具有高精度的测距性能  
High-precision distance measurement performance
- 具有极高的响应速度  
Extremely high response speed

### 1.3 应用场景 / Applications

EDV152主要应用于智能床防夹场景，通过检测人体是否进入床周边危险区域，实现防夹保护功能，保障用户安全。

EDV152 is mainly used in smart bed anti-pinch scenarios. By detecting whether the

human body enters the dangerous area around the bed, it realizes the anti-pinch protection function and ensures user safety.

## 1.4 外形结构 / Appearance Structure

全套产品 / Complete Set:



图 1.1: 产品实物图 / Fig 1.1: Product Photos

单个模块 / Single Module:



图 1.2: 单个模块 / Fig 1.2: Single Module

模块上方有两个插口，正面看：左侧为"输入"，右侧为"输出"。主模块的输出连接到从模块的输入。

There are two ports on the top of the module. Looking from the front: the left is "Input" and the right is "Output". The output of the master module is connected to the input of the slave module.

### 1.4.1 外形尺寸 / Dimension

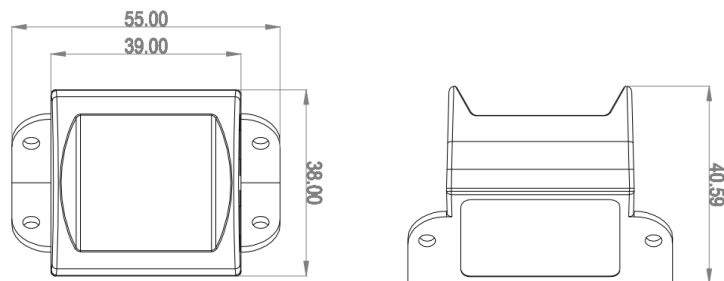


图 1.3: 外形尺寸 / Fig 1.3: Dimensions

| 参数名称<br>Parameter     | 数值<br>Value | 单位<br>Unit |
|-----------------------|-------------|------------|
| 整机长度 / Overall Length | 55          | mm         |
| 整机宽度 / Overall Width  | 38          | mm         |
| 整机高度 / Overall Height | 40.59       | mm         |

表 1.1: 外形尺寸参数 / Tab 1.1: Dimension Parameters

## 1.5 性能参数 / Parameters & Performances

### 1.5.1 参数列表 / Parameters

| 类别<br>Category     | 参数<br>Parameter              | 数值<br>Value | 单位<br>Unit | 说明<br>Remarks |
|--------------------|------------------------------|-------------|------------|---------------|
| 功能参数<br>Functional | 检测帧率<br>Frame Rate           | 20          | Hz         | -             |
| 功能参数<br>Functional | 距离分辨率<br>Distance Resolution | 8           | cm         | -             |
| 电气参数<br>Electrical | 输入电压<br>Input Voltage        | 5           | V          | -             |
| 电气参数<br>Electrical | 工作中心频率<br>Center Frequency   | 60          | GHz        | -             |

表 1.2: 参数表 / Tab 1.2: Parameter Table

接口说明 / Interface Description

2.1 硬件接口 / Hardware Interface

输入端引脚说明（主&从） / Input Pin Description (Master & Slave):

| 引脚<br>Pin | 说明<br>Description         |
|-----------|---------------------------|
| 5V        | 5V电源输入<br>5V Power Supply |
| G         | GND接地<br>Ground           |
| A         | 接对端485 A口<br>RS485 A Port |
| B         | 接对端485 B口<br>RS485 B Port |

表 2.1: 输入端引脚说明 / Tab 2.1: Input Pin Description

输出端引脚说明 / Output Pin Description:

主模块输出口与从模块输入口对接，输出端引脚与输入端相同。  
The master module output port is connected to the slave module input port, and the output pins are the same as the input pins.

2.2 软件接口 / Software Interface

| 接口<br>Interface | 说明<br>Description             |
|-----------------|-------------------------------|
| UART串口<br>UART  | 波特率115200<br>Baud Rate 115200 |

表 2.2: 软件接口 / Tab 2.2: Software Interface

串口协议见：《EDV电动床防夹串口协议》  
See 《EDV Bed Anti-Pinch Serial Protocol》 for detailed protocol.



## 注 / Notes:

1. 床尺寸默认值: 左侧, 180, 200, 30, 20

Default bed dimensions: Left side, 180, 200, 30, 20

2. 设置床尺寸后会自动设置雷达安装高度为床脚高度减10, 如无变化可不用再配置

After setting the bed dimensions, the radar installation height will be automatically set to the bed foot height minus 10. If there is no change, no further configuration is needed

3. 设置成功后, 会写入Flash永久保存, 掉电重启和擦写固件后无需重新设置

After successful setting, it will be written to Flash for permanent storage. No need to reset after power restart or firmware rewrite

## 第三部分 / Chapter 3

# 使用说明 / User Guide

### 3.1 安装指南 / Installation Guide

安装位置：长边与水平面平行，产品侧面安装在床腿，离地高约20cm处，俯视方向看芯片平面与床头平面45度夹角。

Installation position: The long side is parallel to the horizontal plane, the product is installed on the side of the bed leg, about 20cm above the ground, and the chip plane forms a 45-degree angle with the head of the bed when viewed from above.

俯视图 / Top View:

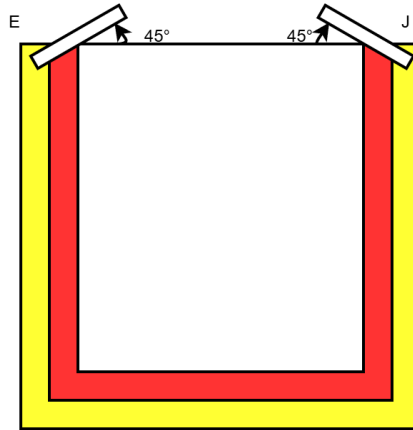


图 3.1: 俯视图 / Fig 3.1: Top View

上图中中间的白色区域为床体，红色区域为围栏检测区域，宽度为30cm。

In the figure above, the white area in the middle is the bed body, and the red area is the fence detection zone with a width of 30cm.

由于雷达的测角和测距都有精度误差，黄色区域为围栏保护区域，宽度为10~15cm。

Due to the precision error of the radar's angle and distance measurement, the yellow area is the fence protection zone with a width of 10~15cm.

侧视图 / Side View:

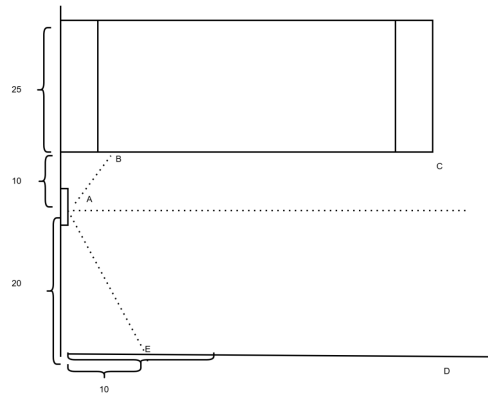


图 3.2: 侧视图 / Fig 3.2: Side View

## 3.2 使用说明 / Usage Instructions

1. 上电启动时间约4秒，上电后默认进入正常探测状态，该状态下，每50ms会进行扫描检测，但如果没有入侵事件或者异常断联发送，每秒上报一次状态；如果有入侵或者主从之间异常断开，会立即上报；

The power-on startup time is about 4 seconds. After power-on, it enters the normal detection state by default. In this state, scanning detection is performed every 50ms. If there is no intrusion event or abnormal disconnection, the status is reported once per second; if there is intrusion or abnormal disconnection between master and slave, it will be reported immediately;

2. 可设置关闭探测，此时主模块会停止发送状态上报消息，从模块为了维持主从连接，每2秒会向主模块通信一次

Detection can be set to off. At this time, the master module will stop sending status report messages, and the slave module will communicate with the master module every 2 seconds to maintain the master-slave connection

3. 发现主从之间断联的时间阈值为5秒

The time threshold for detecting disconnection between master and slave is 5 seconds

4. 检测区域：单手或单脚或差不多大小的物体，单次在检测区域内的运动，检测率可能低于90%，但一般在80%以上。双脚进入或者运动超过1次，检测率大于90%。检测区域宽度为30cm（鞋子的长度）。

Detection zone: For objects the size of a single hand or foot, a single movement in the detection zone may have a detection rate lower than 90%, but generally above 80%. If both feet enter or movement exceeds once, the detection rate is greater than 90%. The detection zone width is 30cm (shoe length).

5. 保护区域：该区域内，有一定概率会被检测到；保护区域以外的临近区域，误触发的概率小于10%。保护区域宽度为10~15cm（视地面和周边反射物情况而定）

Protection zone: In this zone, there is a certain probability of being detected; in the adjacent area outside the protection zone, the probability of false triggering is less than 10%. The protection zone width is 10~15cm (depending on the ground and

surrounding reflectors)